Algebra SEP Worksheet 1 7/18/23

Linear Algebra

Warm Up

1. (August 2019 Problem 4) This problem involves 4×4 matrices with entries in $\mathbb{R}$. We will say that a matrix M is an E**-matrix** if M has 1's along the diagonal and a single nonzero entry off the diagonal. For instance:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 31 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

is an E-matrix. Express the matrix A below as a product of E-matrices.

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$$

2.

(a) Give an example of a finite dimensional vector space V over $\mathbb{C}$, and a nonzero linear transformation $A\colon V \to V$ whose only eigenvalue is 0.

(b) Prove that the only eigenvalue of $A\colon V \to V$ is 0 if and only if A is nilpotent. (V is still f.d. over $\mathbb{C}$.)

3. Give an example of a nonzero finite-dimensional vector space V over $\mathbb{C}$ and an invertible linear transformation $A\colon V \to V$ such that A is not diagonalizable.

Work Out

4. (August 2020 Problem 2) Let M be an $n \times n$ matrix with real entries. Let M^T be its transpose. Each of these two matrices defines a linear transformation $\mathbb{R}^n \to \mathbb{R}^n$; let A denote the transformation corresponding to M, and let B denote that corresponding to M^T.

(a) Prove that any vector in $\ker(A)$ is orthogonal to any vector in $\operatorname{im}(B)$ with respect to the standard inner product (dot product).

(b) Prove that $\mathbb{R}^n$ is the direct sum of $\ker(A)$ and $\operatorname{im}(B)$.

5. (August 2014 Problem 3 (a)-(c))

(a) Let V be a finite-dimensional vector space over $\mathbb{C}$, and let $T\colon V \to V$ be a linear transformation. Show that T has an eigenvector.

(b) Give an example of a nonzero finite-dimensional vector space V over $\mathbb{R}$ and a linear transformation $T\colon V \to V$ such that T does not have an eigenvector.

(c) Does a linear transformation of an infinite-dimensional vector space over $\mathbb{C}$ have to have an eigenvector? Either prove this is the case, or give an example of a linear transformation of an infinite-dimensional vector space with no eigenvector.

6. Let V and W be finite dimensional vector spaces over the same field (for the sake of argument, let's say over $\mathbb{R}$). Let $T\colon V \to V$ and $U\colon W \to W$ be linear maps. Consider the linear map $T \oplus U\colon V \oplus W \to V \oplus W$ which takes (v, w) to (Tv, Uw).

(a) Let $\mathcal{B}_1 = \{v_1, \ldots, v_n\}$ be a basis for V, and $\mathcal{B}_2 = \{w_1, \ldots, w_m\}$ be a basis for W. If A is the matrix of T with respect to $\mathcal{B}_1$, and B is the matrix of U with respect to $\mathcal{B}_2$, what is the matrix of $T \oplus U$ with respect to the basis $\mathcal{B} = \{(v_1, 0), \ldots, (v_n, 0), (0, w_1), \ldots, (0, w_n)\}$?

(b) What are the eigenvalues of $T \oplus U$ in terms of the eigenvalues of T and the eigenvalues of U?

(c) What is the trace of $T \oplus U$ in terms of the trace of T and the trace of U?

(d) Same question for the determinant of $T \oplus U$.

(e) Same question for the characteristic polynomial $\chi_{T \oplus U}$ of $T \oplus U$.

(f) Let $\mu_T(x)$ be the minimal polynomial of T, and $\mu_U(x)$ be the minimal polynomial of U. Prove that the minimal polynomial of $T \oplus U$ is $\operatorname{lcm}(\mu_T, \mu_U)$.

7. (January 1983 Problem 1) Suppose V is a finite dimensional vector space over $\mathbb{C}$, and $T\colon V \to V$ is a linear operator. Suppose the subspace of V spanned by all eigenvectors of T is 1-dimensional. Prove that the minimal polynomial of T is $(x - \alpha)^n$, where $n = \dim V$ and α is some complex number.

8.

(a) Give an example of two linear transformations with the same characteristic polynomial but different minimal polynomials.

(b) Give an example of two linear transformations with the same minimal polynomials which are not similar.

9. (August 2014 Problem 3 (d)) Suppose that T and U are two linear transformations on a finite-dimensional vector space V over $\mathbb{C}$ which satisfy $TU = UT$. Prove that there is some $v \in V$ which is an eigenvector for both T and U.